

IEEE 802.11 Architecture

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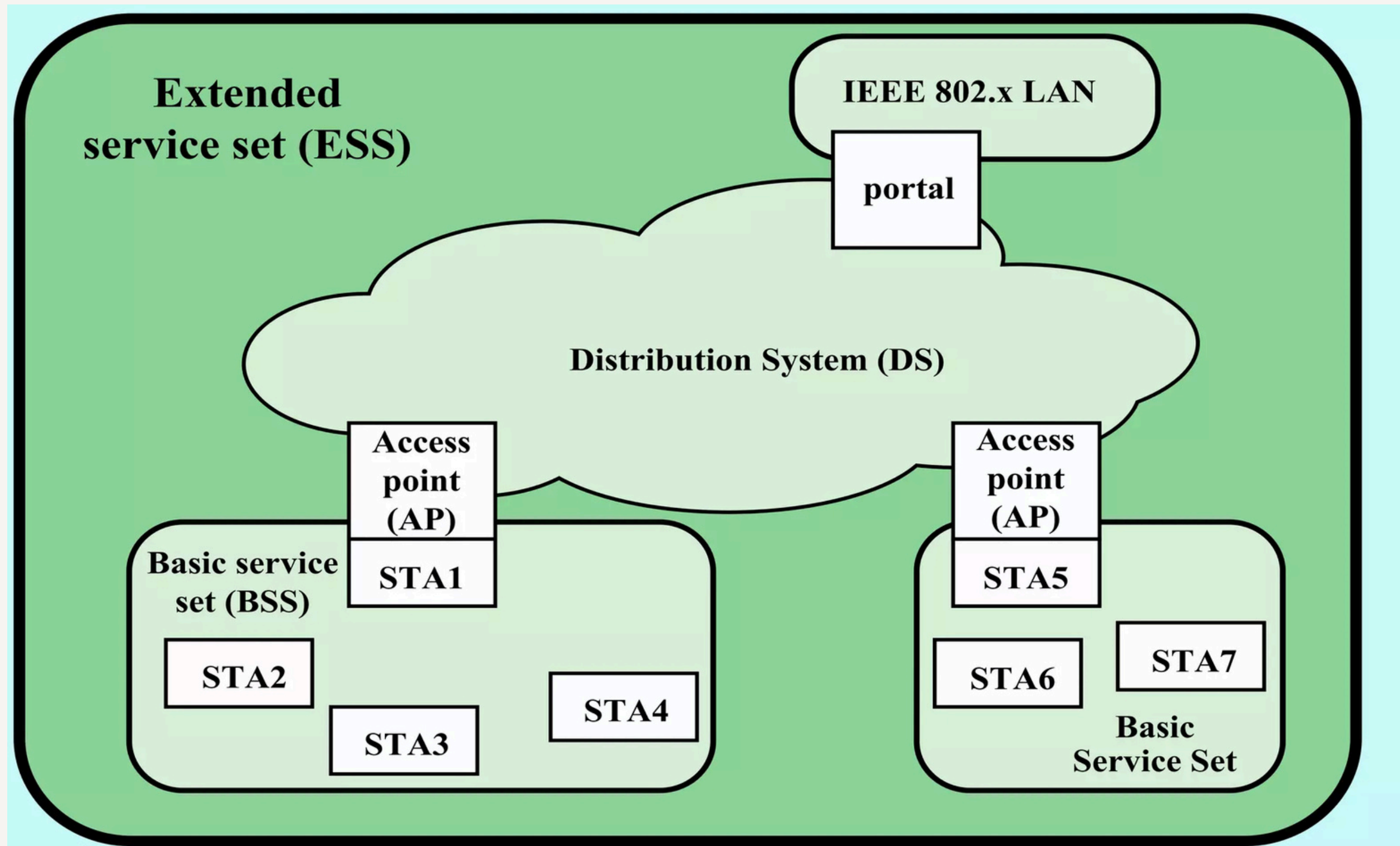
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STA = station

Figure 13.4 IEEE 802.11 Architecture

Basic Service Set (BSS)

- Basic Service Set (BSS) is the smallest unit of WLAN
- Contains wireless stations using the same MAC protocol
- Stations share and compete for the same wireless medium
- BSS may work independently or connect to a DS
- Access Point (AP) is used to connect BSS to DS
- BSS is commonly referred to as a cell

Access Point and Distribution System

- Access Point acts as a bridge and relay
- Stations in a BSS do not communicate directly
- Data is sent from source station to AP first
- AP forwards data to destination station
- Communication to external networks goes via DS
- DS may be wired, switched, or wireless

IBSS and Extended Service Set (ESS)

- IBSS has no Access Point involved
- Stations communicate directly with each other
- IBSS is also called an ad hoc network
- Stations may dynamically join or leave a BSS
- Multiple BSSs can overlap geographically
- ESS connects multiple BSSs as one logical LAN

